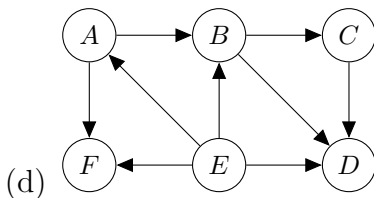
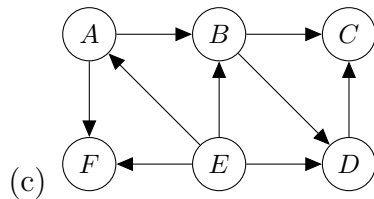
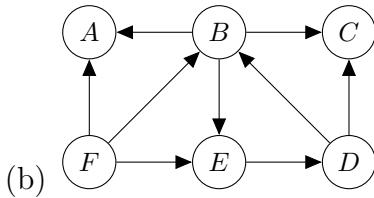
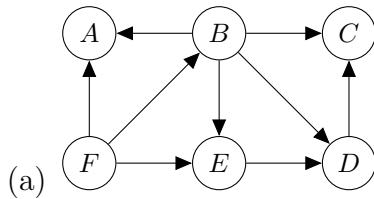
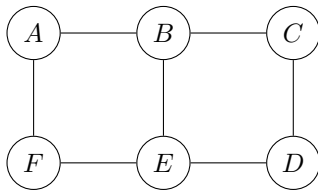


CS 791: Quiz 4

1. Given an MRF, convert it into a minimal BN:-



..2 (multicorrect, -0.5 for each wrong answer)

Correct: (a), (c)

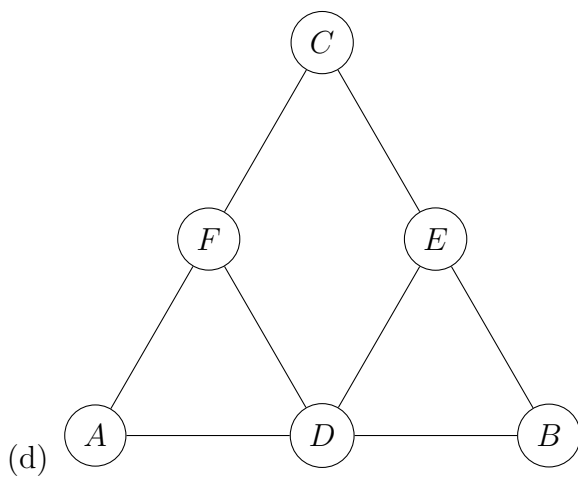
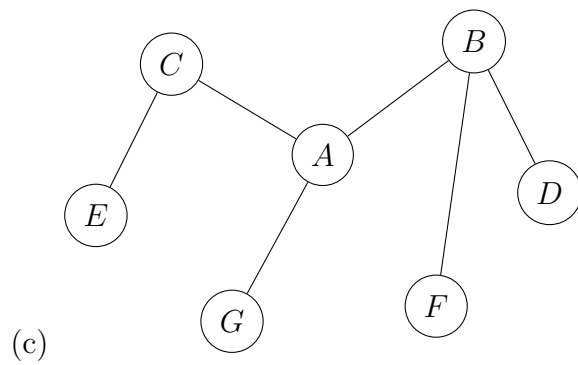
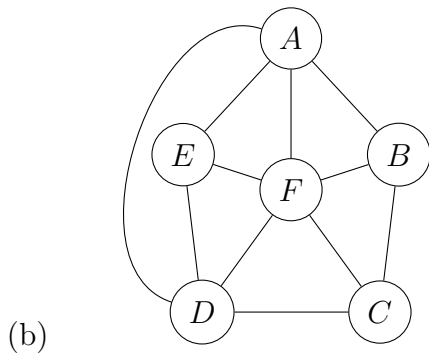
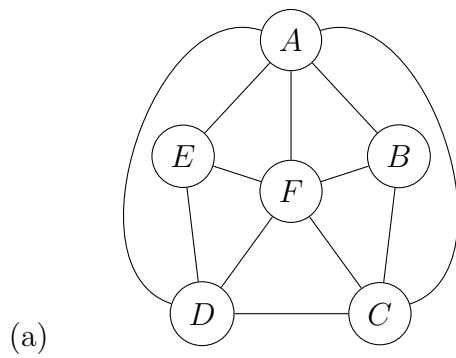
(a) This BN can be made with the variable elimination order - C, D, E, A, B, F

(b) This BN is not a valid BN due to the presence of the cycle - $B \rightarrow E \rightarrow D \rightarrow B$

(c) This BN can be made with the variable elimination order - C, D, B, F, A, E

(d) This BN has extra CIs than MRF. Eg. $E \perp\!\!\!\perp C \mid B$ in the BN but not in the MRF

2. MRF have perfect BN (chordality) Which of the following MRFs have perfect BNs?



..2 (multicorrect, -0.5 for each wrong answer)

Correct: (a), (c)

(a) There is no minimal cycle of length greater than 3.

(b) There is minimal cycle $A-B-C-D-A$ of length 4.

- (c) There is no minimal cycle of length greater than 3.
- (d) There is minimal cycle $D—F—C—E—D$ of length 4.

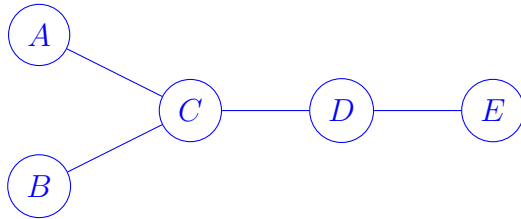
3. Which of the following is true?

- (a) If in an MRF, all the nodes or two non-adjacent nodes are simplicial, then the MRF is chordal.
- (b) Every BN can be converted to a perfect MRF.
- (c) Any MRF can be converted to a perfect BN.
- (d) There can multiple junction trees for the same MRF.

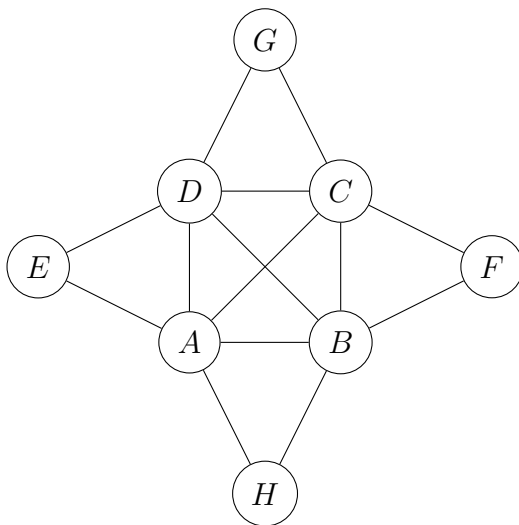
..1

Correct: (d)

- (a) If an MRF is chordal, all the nodes or two non-adjacent nodes are simplicial, but vice versa is not true.
- (b) BNs with immoralities cannot be converted to perfect MRFs.
- (c) Only non chordal MRFs cannot be converted to perfect BNs.
- (d) An MRF for which multiple junction trees exist is:-



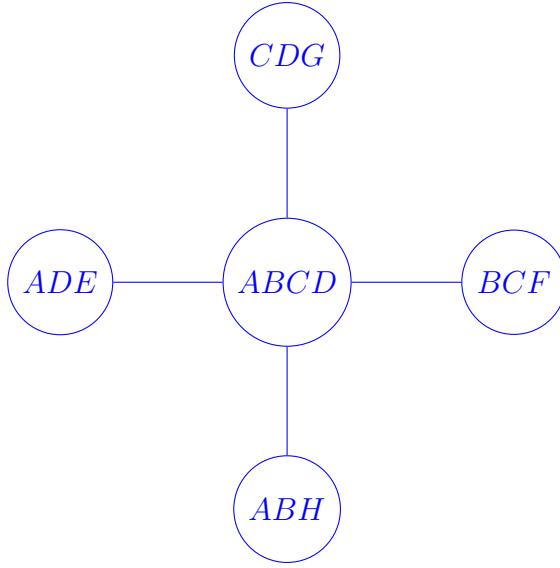
4. Given the MRF



- (a) What are the number of nodes in the above MRF's junction tree, formed by taking the variables according to a simplicial ordering? ..1

Correct: 5

The correct junction tree is:-



(b) Which of the following messages is created according to the message passing algorithm on the above tree?

- i. $m_{ABC \rightarrow ABH}$
- ii. $m_{ABH \rightarrow BCF}$
- iii. $m_{ABCD \rightarrow BCF}$
- iv. $m_{BCDF \rightarrow ABD}$

..1

Correct: iii

The correct messages can be clearly seen from the above junction tree.

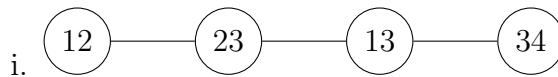
5. Consider the undirected graphical model below over binary variables.

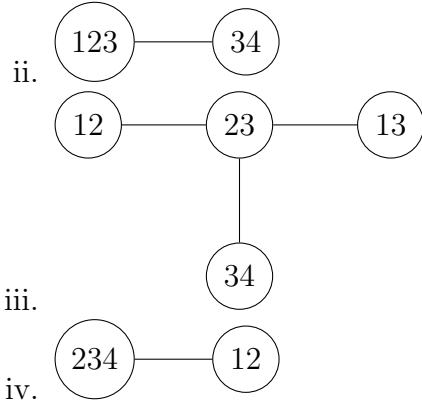
$$\Pr(x_1, x_2, x_3, x_4) = \frac{1}{Z} \psi_a(x_1, x_2) \psi_b(x_2, x_3) \psi_c(x_1, x_3) \psi_d(x_3, x_4)$$

where the potentials are defined as follows.

$\psi_a(x_1, x_2)$			$\psi_b(x_2, x_3)$		
	$x_2 = 0$	$x_2 = 1$		$x_3 = 0$	$x_3 = 1$
$x_1 = 0$	1	2	$x_2 = 0$	4	1
$x_1 = 1$	3	4	$x_2 = 1$	2	3
$\psi_c(x_1, x_3)$			$\psi_d(x_3, x_4)$		
	$x_3 = 0$	$x_3 = 1$		$x_4 = 0$	$x_4 = 1$
$x_1 = 0$	3	4	$x_3 = 0$	2	3
$x_1 = 1$	1	2	$x_3 = 1$	4	1

(a) Which of the following is the correct junction tree?

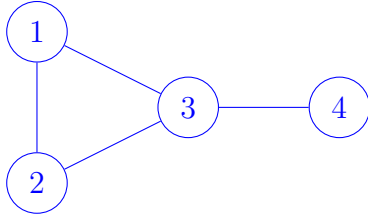




..1

Correct: ii

The MRF for the above distribution is:-



- (b) Let the maximum value of $\Pr(x_1, x_2, x_3, x_4)$ be of the form $\frac{k}{Z}$, where Z is the normalization constant. Find k. ..2

Correct: 96

x_1, x_2, x_3	ψ_{123}				x_3, x_4	ψ_{34}
0, 0, 0	12				0, 0	2
0, 0, 1	4				0, 1	3
0, 1, 0	12	x_3	$m_{123 \rightarrow 34}$	(x_1, x_2) assignments	1, 0	4
0, 1, 1	24	0	12	(0, 0); (0, 1); (1, 0)	1, 1	1
1, 0, 0	12	1	24	(0, 1); (1, 1)		
1, 0, 1	6					
1, 1, 0	8					
1, 1, 1	24					

- (c) How many possible assignments of values of x_1, x_2, x_3, x_4 for which $\Pr(x_1, x_2, x_3, x_4)$ is maximum exist? ..1

Correct: 2

It can be seen from the above tables that the correct assignments of (x_1, x_2, x_3, x_4) are (0,1,1,0) and (1,1,1,0)

Total: 11